

METRIC SPACES: COMPLETE NEW LECTURES

Open Balls, Topology of Open/Closed Sets, Derived Sets, Closures, Interiors & Subspaces

Course Code: MATMJ53 — Lectures by: AP Sir — Platform: www.sciqb.com

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1 Section 1: Open Sets and Fundamental Theorems

1.1 Interior Points and Open Sets

Definition: Interior Point and Open Set

Let (X, d) be a metric space and $G \subseteq X$.

- A point $x \in G$ is called an **interior point** of G if and only if there exists an open sphere $S_r(x)$ centered at x with radius $r > 0$ such that:

$$S_r(x) \subseteq G \quad (1)$$

- A subset $G \subseteq X$ is called an **open set** in (X, d) if every point of G is an interior point of G .

1.2 Fundamental Theorems on Open Sets

Theorem 1: The Empty Set $\emptyset$ and Full Space X are Open

Theorem: In any metric space (X, d) , the empty set $\emptyset$ and the full space X are open sets.

Proof:

1. **For $\emptyset$:** Since $\emptyset$ contains no points, the condition that every point of $\emptyset$ is an interior point holds vacuously. Hence, $\emptyset$ is open.
2. **For X :** Let $x \in X$ be arbitrary. For any $r > 0$, the open sphere $S_r(x) = \{y \in X \mid d(y, x) < r\} \subseteq X$. Thus, every point in X is an interior point of X , so X is open. $\square$

Theorem 2: Every Open Sphere is an Open Set

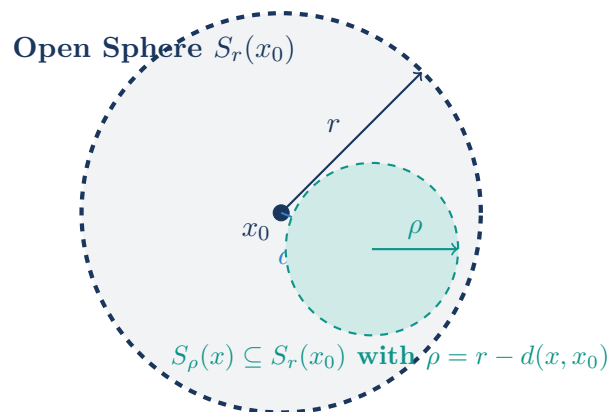
Theorem: In a metric space (X, d) , every open sphere $S_r(x_0)$ is an open set.

Proof:

1. Let $S_r(x_0)$ be an open sphere centered at $x_0 \in X$ with radius $r > 0$.
2. Let $x \in S_r(x_0)$ be an arbitrary point.
3. By definition of the open sphere, $d(x, x_0) < r \implies \rho = r - d(x, x_0) > 0$.
4. **Claim:** The open sphere $S_\rho(x) \subseteq S_r(x_0)$.
5. To prove this claim, let $y \in S_\rho(x) \implies d(y, x) < \rho$.
6. By the triangle inequality [M4] on (X, d) :

$$\begin{aligned}
 d(y, x_0) &\leq d(y, x) + d(x, x_0) \\
 &< \rho + d(x, x_0) \\
 &= (r - d(x, x_0)) + d(x, x_0) = r
 \end{aligned} \tag{2}$$

7. Since $d(y, x_0) < r$, we have $y \in S_r(x_0)$.
8. Therefore, $S_\rho(x) \subseteq S_r(x_0)$.
9. Since $x \in S_r(x_0)$ was arbitrary, every point of $S_r(x_0)$ is an interior point.
10. Hence, $S_r(x_0)$ is an **open set**. □



Theorem 3: Arbitrary Union of Open Sets is Open

Theorem: In a metric space (X, d) , the union of an arbitrary collection of open sets is an open set.

Proof:

1. Let $\{G_\lambda \mid \lambda \in \Lambda\}$ be an arbitrary family of open sets in (X, d) .
2. Let $G = \bigcup_{\lambda \in \Lambda} G_\lambda$.
3. If $G = \emptyset$, then G is open by Theorem 1.
4. If $G \neq \emptyset$, let $x \in G$.
5. By definition of the union, $x \in G_{\lambda_0}$ for some specific $\lambda_0 \in \Lambda$.
6. Since G_{λ_0} is open, there exists $r > 0$ such that $S_r(x) \subseteq G_{\lambda_0}$.
7. Since $G_{\lambda_0} \subseteq \bigcup_{\lambda \in \Lambda} G_\lambda = G$, we have $S_r(x) \subseteq G$.
8. Thus every point of G is an interior point of G , proving that G is **open**. □

Theorem 4: Finite Intersection of Open Sets is Open

Theorem: In a metric space (X, d) , the intersection of a finite number of open sets is an open set.

Proof:

1. Let $G_1, G_2, \dots, G_n$ be a finite collection of open sets in (X, d) .
2. Let $G = \bigcap_{i=1}^n G_i$.
3. If $G = \emptyset$, G is open. If $G \neq \emptyset$, let $x \in G$.
4. Then $x \in G_i$ for each $i \in \{1, 2, \dots, n\}$.
5. Since each G_i is open, there exist radii $r_1, r_2, \dots, r_n > 0$ such that $S_{r_i}(x) \subseteq G_i$ for each i .
6. Define $r = \min\{r_1, r_2, \dots, r_n\}$. Since this is the minimum of finitely many positive numbers, $r > 0$.
7. Then $S_r(x) \subseteq S_{r_i}(x) \subseteq G_i$ for all $i = 1, 2, \dots, n$.
8. Consequently, $S_r(x) \subseteq \bigcap_{i=1}^n G_i = G$.
9. Hence x is an interior point of G , which proves G is **open**. □

Important Counterexample: Infinite Intersection of Open Sets May NOT be Open

In $(\mathbb{R}, |\cdot|)$, consider the family of open intervals $G_n = \left(-\frac{1}{n}, \frac{1}{n}\right)$, $n \in \mathbb{N}$.

$$\bigcap_{n=1}^{\infty} G_n = \bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\} \quad (3)$$

The singleton set $\{0\}$ is **closed** and **not open** in $\mathbb{R}$ because $\forall r > 0, S_r(0) = (-r, r) \not\subseteq \{0\}$.

1.3 Discrete Metric Space

Definition: Discrete Metric Space

Let X be any arbitrary non-empty set. The **discrete metric** on X is defined as:

$$d(x, y) = 0 \quad \text{if } x = y, \quad \text{and} \quad d(x, y) = 1 \quad \text{if } x \neq y \quad (4)$$

Key Topological Properties of Discrete Metric Spaces:

- For any point $x_0 \in X$ and radius $r \leq 1$, $S_r(x_0) = \{x_0\}$. Thus, **every singleton set is an open ball**, and hence open.
- Since every subset is a union of singletons, **every subset in a discrete metric space is both open and closed (clopen)**.

2 Section 2: Closed Sets and Derived Sets

2.1 Closed Sets in Metric Spaces

Definition: Closed Set

Let (X, d) be a metric space. A subset $F \subseteq X$ is called a **closed set** if and only if its complement $F^c = X \setminus F$ is an open set in (X, d) .

Examples of Closed Sets

1. **Singleton Sets in $(\mathbb{R}, |\cdot|)$:** For any $x \in \mathbb{R}$, the singleton $\{x\}$ is closed because its complement:

$$\{x\}^c = (-\infty, x) \cup (x, \infty)$$

is the union of two open intervals (open sets), hence open. Thus, **every singleton set in a metric space is closed**.

2. **Closed Intervals in $(\mathbb{R}, |\cdot|)$:** The interval $[a, b]$ is closed because $[a, b]^c = (-\infty, a) \cup (b, \infty)$ is open.

2.2 Topological Properties of Closed Sets

Duality Properties of Closed Sets

1. $\emptyset$ and X are closed sets.
2. The **arbitrary intersection** of closed sets is closed: $(\bigcap_{\lambda} F_{\lambda})^c = \bigcup_{\lambda} F_{\lambda}^c$ (open).
3. The **finite union** of closed sets is closed: $(\bigcup_{i=1}^n F_i)^c = \bigcap_{i=1}^n F_i^c$ (open).

2.3 Limit Point Characterization of Closed Sets

Theorem: Closed Sets Contain All Their Limit Points

Theorem: In a metric space (X, d) , a subset $F \subseteq X$ is closed if and only if F contains all its limit points ($F' \subseteq F$).

Proof:

1. ($\implies$) **Necessary Condition:** Assume F is closed $\implies F^c$ is open.

- If $F' = \emptyset$, then $F' \subseteq F$ holds trivially.
- If $F' \neq \emptyset$, let $p \in F'$. We must show $p \in F$.
- Suppose for contradiction that $p \notin F \implies p \in F^c$.
- Since F^c is open, $\exists r > 0$ such that $S_r(p) \subseteq F^c = X \setminus F$.
- This implies $S_r(p) \cap F = \emptyset \implies (S_r(p) \setminus \{p\}) \cap F = \emptyset$.
- This contradicts the fact that p is a limit point of F . Hence $p \in F$, so $F' \subseteq F$.

2. ($\impliedby$) **Sufficient Condition:** Assume $F' \subseteq F$. We must show F^c is open.

- Let $x \in F^c = X \setminus F$. Then $x \notin F \implies x \notin F'$ (since $F' \subseteq F$).
- Since x is not a limit point of F , there exists $r > 0$ such that:

$$(S_r(x) \setminus \{x\}) \cap F = \emptyset$$

- Since $x \notin F$, this gives $S_r(x) \cap F = \emptyset \implies S_r(x) \subseteq F^c$.
- Thus every point of F^c is an interior point of F^c , so F^c is open $\implies F$ is **closed**. $\square$

2.4 Theorem: Every Derived Set is Closed

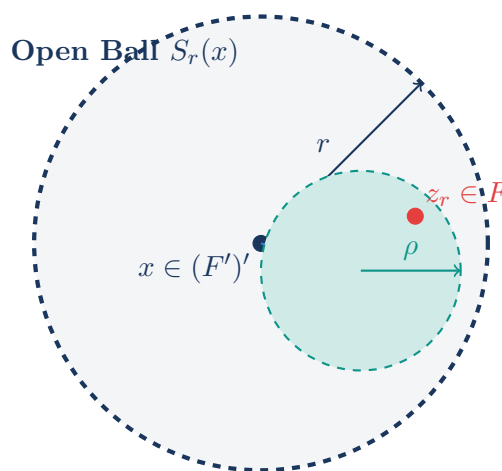
Theorem: The Derived Set F' is Closed in (X, d)

Theorem: Let (X, d) be a metric space and $F \subseteq X$. Then the derived set F' is a closed set in (X, d) .

Proof:

1. We show that $(F')' \subseteq F'$, which proves F' is closed.
2. If $(F')' = \emptyset$, $(F')' \subseteq F'$ is trivial.
3. Let $x \in (F')'$ (i.e., x is a limit point of F'). We must prove $x \in F'$ (i.e., x is a limit point of F).
4. Let $S_r(x)$ be any open sphere centered at x with radius $r > 0$.
5. Since $x \in (F')'$, there exists a point $y_r \in F'$ distinct from x with $y_r \in S_r(x)$.
6. Thus $d(x, y_r) < r \implies \rho = r - d(x, y_r) > 0$.
7. Consider the open sphere $S_\rho(y_r)$. By Theorem 2, $S_\rho(y_r) \subseteq S_r(x)$.
8. Since $y_r \in F'$ (y_r is a limit point of F), the sphere $S_\rho(y_r)$ contains a point $z_r \in F$ distinct from y_r :

$$z_r \in (S_\rho(y_r) \setminus \{y_r\}) \cap F \quad (5)$$
9. Since $z_r \in S_\rho(y_r) \subseteq S_r(x)$, we have $z_r \in S_r(x)$ and $z_r \in F$.
10. By choosing $\rho_1 = \min(\rho, d(x, y_r)) > 0$, we strictly ensure $z_r \neq x$.
11. Therefore, $(S_r(x) \setminus \{x\}) \cap F \neq \emptyset$.
12. This proves that x is a limit point of $F \implies x \in F'$.
13. Hence $(F')' \subseteq F'$, which establishes that F' is a **closed set**. □



Point $z_r \in F$ lies inside $S_\rho(y_r) \subseteq S_r(x)$, proving $x \in F'$

3 Section 3: Closure of a Set and Kuratowski Properties

3.1 Definition of Closure

Definition: Closure of a Set ($\bar{A}$)

Let (X, d) be a metric space and $A \subseteq X$. The **closure** of A , denoted by $\bar{A}$ or $\text{Cl}(A)$, is defined as:

$$\bar{A} = A \cup A' \quad (6)$$

A point $x \in \bar{A}$ is called an **adherent point** of A ($\forall r > 0, S_r(x) \cap A \neq \emptyset$).

Theorem: 7 Fundamental Properties of the Closure Operator

Let (X, d) be a metric space and $A, B \subseteq X$:

- (i) $\bar{A}$ is a **closed set** in (X, d) .
- (ii) **Monotonicity:** $A \subseteq B \implies \bar{A} \subseteq \bar{B}$.
- (iii) **Characterization:** A is closed $\iff \bar{A} = A$.
- (iv) **Minimality:** $\bar{A}$ is the **smallest closed set** containing A .
- (v) **Finite Additivity:** $\overline{A \cup B} = \bar{A} \cup \bar{B}$.
- (vi) **Intersection Sub-additivity:** $\overline{A \cap B} \subseteq \bar{A} \cap \bar{B}$.
- (vii) **Idempotence:** $\overline{\bar{A}} = \bar{A}$.

3.2 Complete Rigorous Proofs for Closure Theorems

Proof of Property (i): $\bar{A}$ is a Closed Set

To Prove: $(\bar{A})^c$ is open.

1. $(\bar{A})^c = (A \cup A')^c = A^c \cap (A')^c$.
2. Let $x \in (\bar{A})^c = A^c \cap (A')^c \implies x \in A^c$ and $x \in (A')^c$.
3. Since $x \notin A'$, x is not a limit point of $A \implies \exists r_1 > 0$ such that $(S_{r_1}(x) \setminus \{x\}) \cap A = \emptyset$.
Since $x \notin A$, $S_{r_1}(x) \cap A = \emptyset \implies S_{r_1}(x) \subseteq A^c$.
4. Since A' is closed, $(A')^c$ is open $\implies \exists r_2 > 0$ such that $S_{r_2}(x) \subseteq (A')^c$.
5. Set $r^* = \min(r_1, r_2) > 0$. Then $S_{r^*}(x) \subseteq A^c \cap (A')^c = (\bar{A})^c$.
6. Thus $(\bar{A})^c$ is open, so $\bar{A}$ is **closed**. □

Proof of Properties (ii), (iii), and (iv)

Proof of (ii): Let $x \in A' \implies \forall r > 0, (S_r(x) \setminus \{x\}) \cap A \neq \emptyset$. Since $A \subseteq B$, $(S_r(x) \setminus \{x\}) \cap B \neq \emptyset \implies x \in B'$. Thus $A' \subseteq B'$. Hence $\overline{A} = A \cup A' \subseteq B \cup B' = \overline{B}$. $\square$

Proof of (iii): A is closed $\iff A' \subseteq A \iff \overline{A} = A \cup A' = A$. $\square$

Proof of (iv): $\overline{A}$ is closed and $A \subseteq \overline{A}$. If F is any closed set with $A \subseteq F$, then $\overline{A} \subseteq \overline{F} = F$. Hence $\overline{A} = \bigcap \{F \supseteq A \mid F \text{ is closed}\}$. $\square$

Proof of Properties (v), (vi), and (vii)

Proof of (v) $[\overline{A \cup B} = \overline{A} \cup \overline{B}]$:

1. $A \subseteq A \cup B \Rightarrow \overline{A} \subseteq \overline{A \cup B}$, and $B \subseteq A \cup B \Rightarrow \overline{B} \subseteq \overline{A \cup B} \Rightarrow \overline{A} \cup \overline{B} \subseteq \overline{A \cup B}$.
2. Also $A \subseteq \overline{A}, B \subseteq \overline{B} \Rightarrow A \cup B \subseteq \overline{A} \cup \overline{B}$. Since $\overline{A} \cup \overline{B}$ is closed, $\overline{A \cup B} \subseteq \overline{A} \cup \overline{B}$.
3. Thus $\overline{A \cup B} = \overline{A} \cup \overline{B}$. □

Proof of (vi) $[\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}]$:

1. $A \cap B \subseteq A \Rightarrow \overline{A \cap B} \subseteq \overline{A}$ and $A \cap B \subseteq B \Rightarrow \overline{A \cap B} \subseteq \overline{B} \Rightarrow \overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$. □
2. **Counterexample for Strict Inclusion:** For $A = (0, 1), B = (1, 2)$ in $\mathbb{R}$:

$$\overline{A \cap B} = \overline{\emptyset} = \emptyset, \quad \text{while } \overline{A} \cap \overline{B} = [0, 1] \cap [1, 2] = \{1\} \neq \emptyset$$

Proof of (vii) $[\overline{\overline{A}} = \overline{A}]$:

1. Since $\overline{A}$ is closed (Property (i)), by Property (iii), $\overline{\overline{A}} = \overline{A}$. □

3.3 Perfect Sets

Definition: Perfect Set

A subset $A \subseteq X$ is called a **perfect set** if and only if A is closed and every point of A is a limit point of A :

$$A = A' \tag{7}$$

Equivalently, a perfect set is a closed set with **no isolated points**.

Examples of Perfect Sets

1. Closed intervals $[a, b]$ in $(\mathbb{R}, |\cdot|)$ are perfect because $[a, b]' = [a, b]$.
2. The whole Euclidean space $\mathbb{R}^n$ is perfect ($(\mathbb{R}^n)' = \mathbb{R}^n$).
3. The Cantor Ternary Set $\mathcal{C} \subset [0, 1]$ is a compact, uncountably infinite perfect set of Lebesgue measure 0.
4. Singleton sets $\{x\}$ are closed but **not** perfect since $\{x\}' = \emptyset \neq \{x\}$.

Perfect Set: $A = [a, b] \Rightarrow A' = [a, b] = A$



Not Perfect: $A = [a, b] \cup \{c\} \Rightarrow A' = [a, b] \neq A$



4 Section 4: The Interior Operator and Topological Boundary

4.1 Definition and Properties of Interior

Definition: Interior Operator (A° / $\text{Int}(A)$)

Let (X, d) be a metric space and $A \subseteq X$. The **interior** of A is:

$$A^\circ = \text{Int}(A) = \{x \in A \mid \exists r > 0 \text{ such that } S_r(x) \subseteq A\} \quad (8)$$

Theorem: Fundamental Properties of the Interior Operator

- (i) $A^\circ \subseteq A$.
- (ii) A° is an **open set** in (X, d) .
- (iii) A is open $\iff A^\circ = A$.
- (iv) If $B \subseteq A$ and B is open, then $B \subseteq A^\circ$.
- (v) A° is the **largest open set** contained in A ($A^\circ = \bigcup \{G \subseteq A \mid G \text{ is open}\}$).
- (vi) $(A \cap B)^\circ = A^\circ \cap B^\circ$ and $A^\circ \cup B^\circ \subseteq (A \cup B)^\circ$.
- (vii) $(A^\circ)^\circ = A^\circ$ (Idempotence).

Proofs of Interior Theorems

Proof of (ii) [A° is Open]:

- Let $x \in A^\circ \implies \exists r > 0$ such that $S_r(x) \subseteq A$.
- We show $S_r(x) \subseteq A^\circ$. Let $y \in S_r(x)$.
- By Theorem 2, $\exists \rho = r - d(x, y) > 0$ such that $S_\rho(y) \subseteq S_r(x) \subseteq A$.
- This shows y is an interior point of $A \implies y \in A^\circ$.
- Thus $S_r(x) \subseteq A^\circ$, which proves x is an interior point of A° . Hence A° is open. $\square$

Proof of (iii) [A is open $\iff A^\circ = A$]:

- If A is open, every $x \in A$ is an interior point $\implies A \subseteq A^\circ \implies A^\circ = A$.
- Conversely, if $A = A^\circ$, since A° is open (by (ii)), A is open. $\square$

Proof of (iv) & (v) [A° is largest open set in A]:

- Let $B \subseteq A$ with B open. Let $x \in B \implies \exists r > 0, S_r(x) \subseteq B \subseteq A \implies x \in A^\circ$.
- Hence $B \subseteq A^\circ$, so $A^\circ = \bigcup \{G \subseteq A \mid G \text{ is open}\}$. $\square$

Duality Relations and Topological Boundary

$$(A^c)^\circ = (\overline{A})^c, \quad \overline{A^c} = (A^\circ)^c, \quad \partial A = \text{Bd}(A) = \overline{A} \setminus A^\circ = \overline{A} \cap \overline{A^c} \quad (9)$$

5 Section 5: Metric Subspaces and Relative Topologies

5.1 Definition of Metric Subspace

Definition: Metric Subspace (Y, d_Y)

Let (X, d) be a metric space, and let Y be a non-empty subset of X .

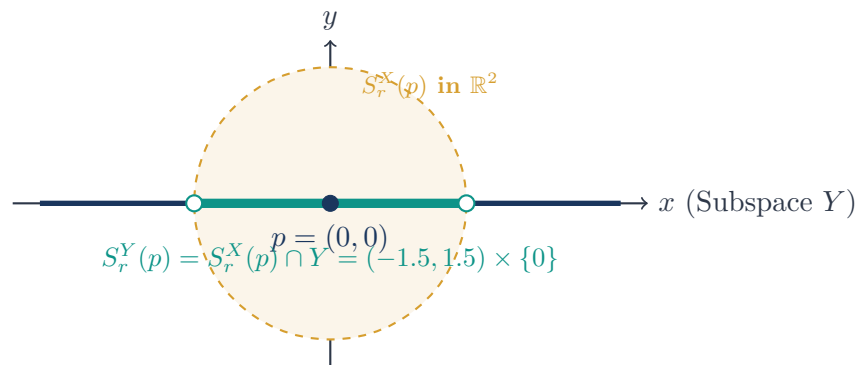
- Define $d_Y : Y \times Y \rightarrow \mathbb{R}$ by $d_Y(x, y) = d(x, y)$ for all $x, y \in Y$ (metric restriction).
- The pair (Y, d_Y) is called a **Metric Subspace** of (X, d) .
- **Subspace Open Sphere:** For any $y_0 \in Y$ and $r > 0$:

$$S_r^Y(y_0) = \{y \in Y \mid d_Y(y, y_0) < r\} = S_r^X(y_0) \cap Y \quad (10)$$

Example: Relative Openness vs. Global Openness

Let $(X, d) = (\mathbb{R}^2, d_2)$ and $Y = \{(x, 0) \in \mathbb{R}^2 \mid x \in \mathbb{R}\}$ (the x -axis). Let $E = (-1, 1) \times \{0\}$.

- In the subspace (Y, d_Y) , $E = S_1^Y((0, 0))$ is an open ball $\implies E$ is **open relative to Y** .
- In the full space $\mathbb{R}^2$, every 2D open disk $S_r^{\mathbb{R}^2}((0, 0))$ contains points with non-zero y -coordinates $\implies S_r^{\mathbb{R}^2}((0, 0)) \not\subseteq E \implies E$ is **NOT open in $\mathbb{R}^2$** .



Relative Open Ball: Open in subspace Y , but 1D line has no interior in $\mathbb{R}^2$

5.2 Fundamental Theorems on Subspace Topology

Theorem: Characterization of Relative Open Sets

Theorem: Let (Y, d) be a subspace of (X, d) . A subset $E \subseteq Y$ is **open relative to Y** if and only if there exists an open set $G \subseteq X$ such that:

$$E = Y \cap G \quad (11)$$

Proof:

1. ($\implies$) Assume E is open in Y .

- For each $p \in E$, $\exists r_p > 0$ such that $S_{r_p}^Y(p) \subseteq E$.
- Note $S_{r_p}^Y(p) = S_{r_p}^X(p) \cap Y = V_p \cap Y$, where $V_p = S_{r_p}^X(p)$ is open in X .
- Let $G = \bigcup_{p \in E} V_p$. Since each V_p is open in X , G is an open set in X .
- Then $Y \cap G = Y \cap \left(\bigcup_{p \in E} V_p \right) = \bigcup_{p \in E} (Y \cap V_p) = \bigcup_{p \in E} S_{r_p}^Y(p) = E$.
- Thus $E = Y \cap G$ for the open set $G \subseteq X$.

2. ($\impliedby$) Assume $E = Y \cap G$ for some open set $G \subseteq X$.

- Let $p \in E = Y \cap G \implies p \in Y$ and $p \in G$.
- Since G is open in X , $\exists r > 0$ such that $S_r^X(p) \subseteq G$.
- Then $S_r^Y(p) = S_r^X(p) \cap Y \subseteq G \cap Y = E$.
- Thus p is an interior point of E in Y , so E is **open relative to Y** . □

Theorem: Characterization of Relative Closed Sets

Theorem: Let (Y, d) be a subspace of (X, d) . A subset $K \subseteq Y$ is **closed relative to Y** if and only if there exists a closed set $F \subseteq X$ such that:

$$K = Y \cap F \quad (12)$$

Proof:

1. K is closed in $Y \iff Y \setminus K$ is open in Y .
2. By the Relative Open Set Theorem, $Y \setminus K = Y \cap G$ for some open set $G \subseteq X$.
3. Taking complements in Y :

$$K = Y \setminus (Y \setminus K) = Y \setminus (Y \cap G) = Y \cap (X \setminus G) = Y \cap F \quad (13)$$

where $F = X \setminus G = G^c$ is a **closed set in X** .

4. Conversely, if $K = Y \cap F$ for closed $F \subseteq X$, then $Y \setminus K = Y \cap (X \setminus F) = Y \cap G$ (where $G = F^c$ is open), proving K is closed in Y . □

Corollaries: Inherited Openness and Closedness

- If Y is **open** in X , then $E \subseteq Y$ is open in $Y \iff E$ is open in X .
- If Y is **closed** in X , then $K \subseteq Y$ is closed in $Y \iff K$ is closed in X .

6 Section 6: Quick Revision Cheat Sheet on New Topics

1. Open Sets, Spheres, and Derived Sets

- **Open Sphere** $S_r(x_0)$: $S_r(x_0) = \{x \in X \mid d(x, x_0) < r\}$.
- **Fundamental Ball Theorem**: Every open sphere $S_r(x_0)$ is an **open set** (radius $\rho = r - d(x, x_0) > 0$).
- **Open Set Properties**:
 - $\emptyset$ and X are open sets.
 - Arbitrary union $\bigcup_{\lambda \in \Lambda} G_\lambda$ is open.
 - Finite intersection $\bigcap_{i=1}^n G_i$ is open (Infinite intersection may fail: $\bigcap_{n=1}^\infty (-1/n, 1/n) = \{0\}$).
- **Derived Set** A' : Set of all limit points of A . **Theorem**: A' is always **closed** in (X, d) ($((A')') \subseteq A'$).

2. Closed Sets, Closures, and Perfect Sets

- **Closed Set Characterization**: F is closed $\iff F^c$ is open $\iff F' \subseteq F \iff \overline{F} = F$.
- **Closure Operator** $\overline{A} = A \cup A'$:
 - $\overline{A}$ is closed; $A \subseteq B \implies \overline{A} \subseteq \overline{B}$; $\overline{\overline{A}} = \overline{A}$.
 - $\overline{A}$ is the **smallest closed set** containing A .
 - $\overline{A \cup B} = \overline{A} \cup \overline{B}$, and $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$.
- **Perfect Set**: A is closed with no isolated points $\iff A = A'$ (e.g., $[a, b]$, $\mathbb{R}^n$, Cantor Set $\mathcal{C}$).

3. Interior Operator, Boundary, and Metric Subspaces

- **Interior Operator** $A^\circ = \text{Int}(A)$:
 - $A^\circ \subseteq A$, and A° is an open set.
 - A is open $\iff A^\circ = A$.
 - A° is the **largest open set** contained in A .
 - $(A \cap B)^\circ = A^\circ \cap B^\circ$, and $(A^\circ)^\circ = A^\circ$.
- **Duality & Boundary**: $(A^c)^\circ = (\overline{A})^c$, $\overline{A^c} = (A^\circ)^c$, $\partial A = \overline{A} \setminus A^\circ = \overline{A} \cap \overline{A^c}$.
- **Metric Subspaces** (Y, d_Y) where $d_Y = d|_{Y \times Y}$:
 - Subspace Ball: $S_r^Y(y) = S_r^X(y) \cap Y$.
 - **Relative Open Set Theorem**: $E \subseteq Y$ is open in $Y \iff E = Y \cap G$ for some open set $G \subseteq X$.
 - **Relative Closed Set Theorem**: $K \subseteq Y$ is closed in $Y \iff K = Y \cap F$ for some closed set $F \subseteq X$.